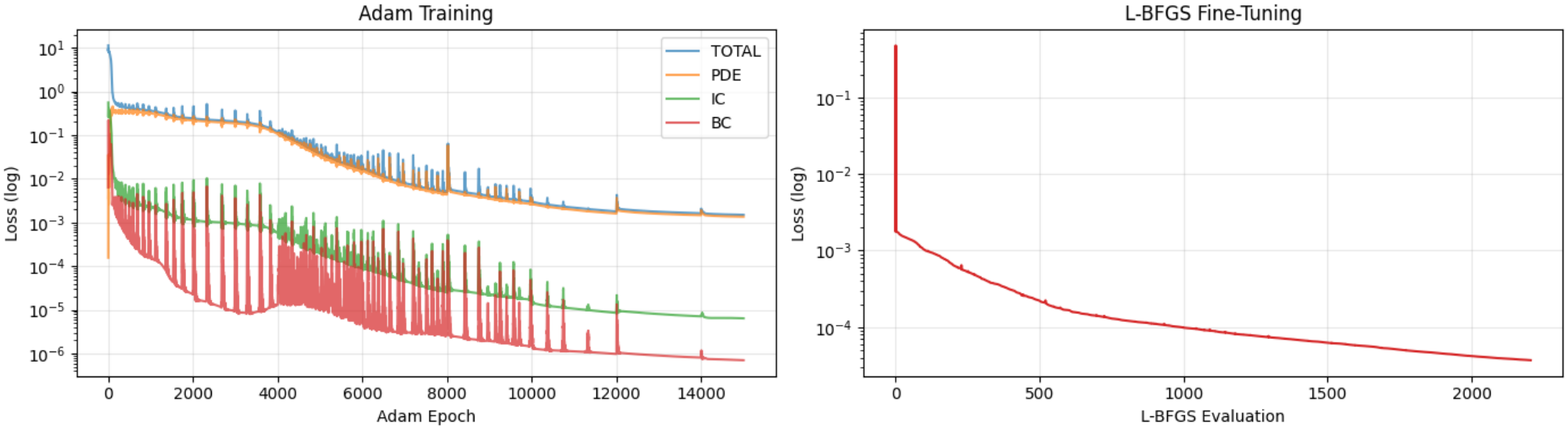
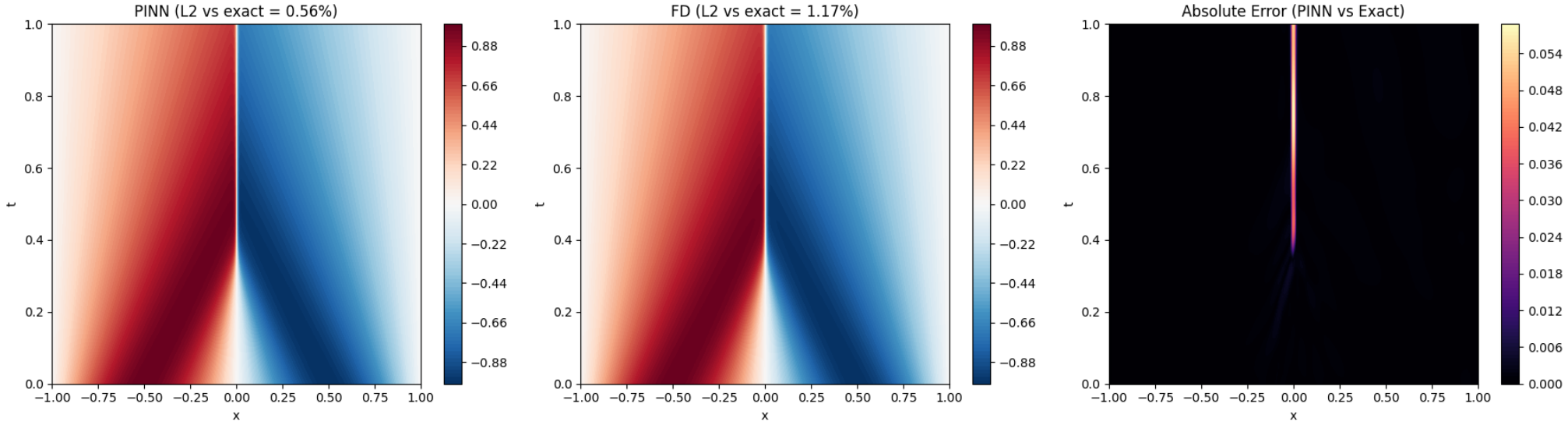
Figure 1. Forward PINN training loss



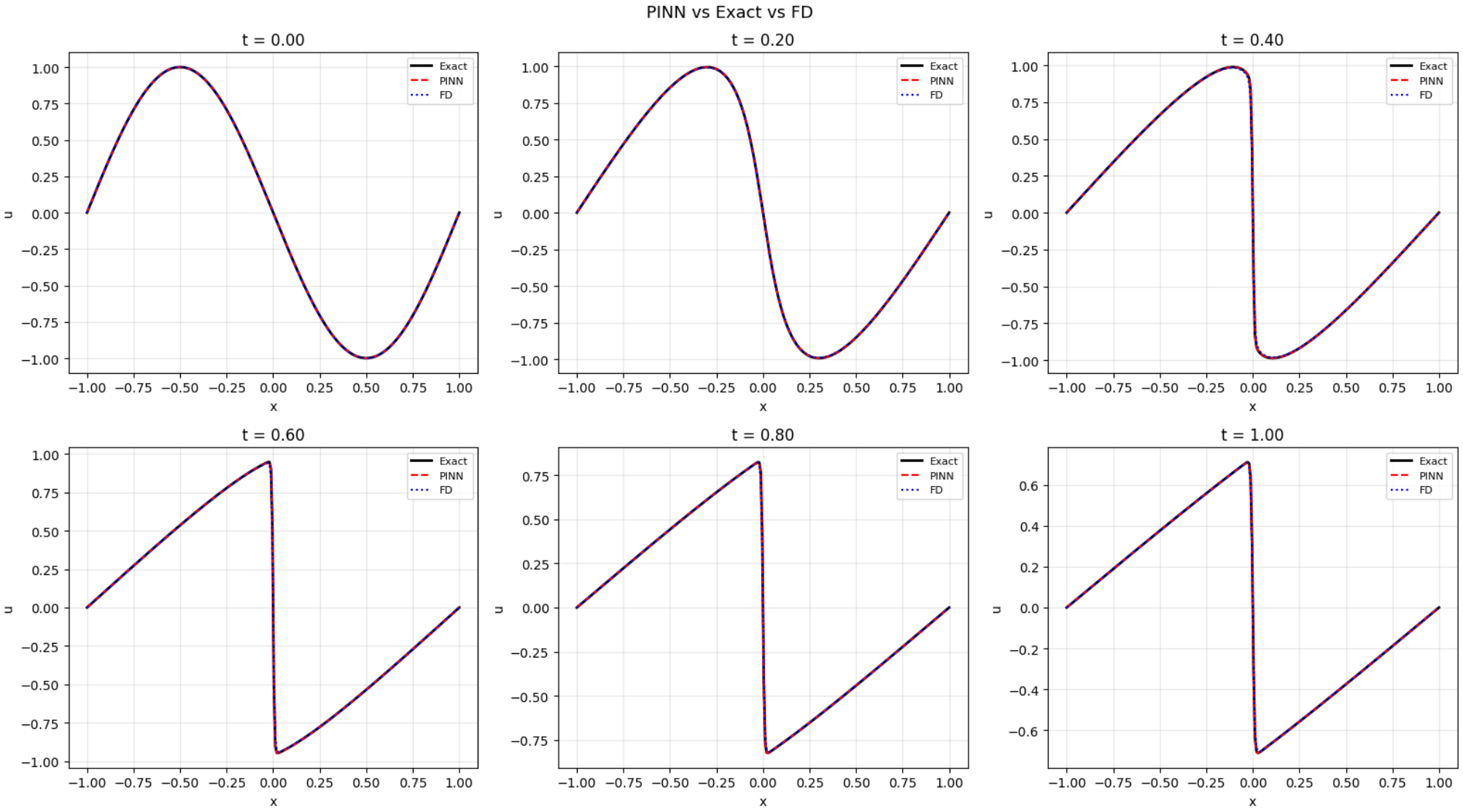
Left: Adam (15,000 epochs, log scale) with total, PDE, IC and BC terms. Right: L-BFGS fine-tuning (100 steps).

Figure 2. Forward solution u(x,t)



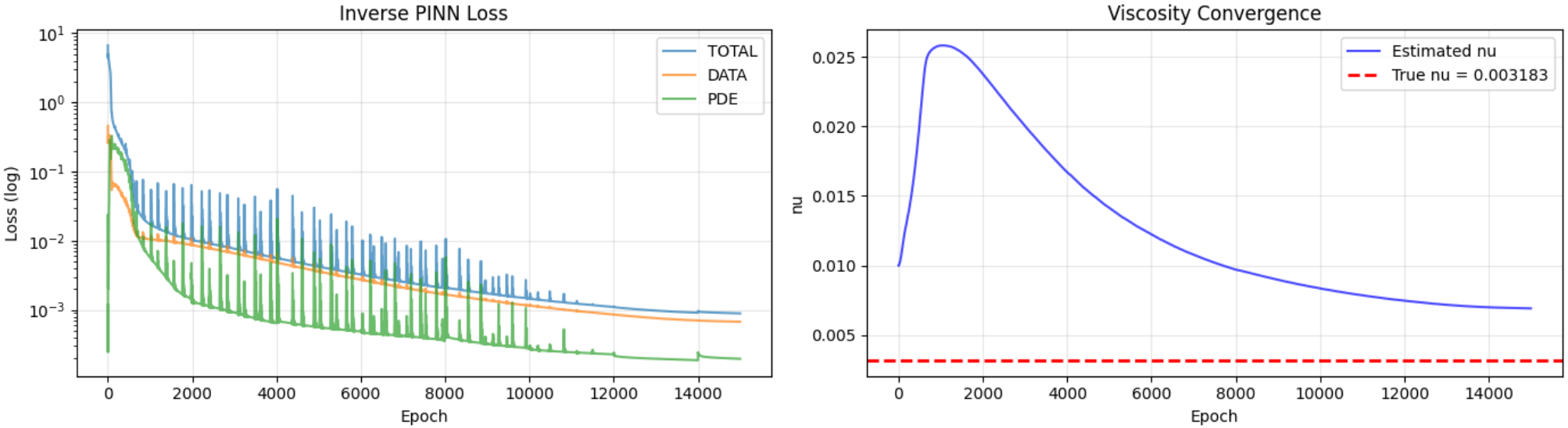
PINN (0.56% rel. L2 vs exact), finite-difference reference (1.17%), and absolute error of PINN vs exact.

Figure 3. PINN vs Exact vs FD at selected times



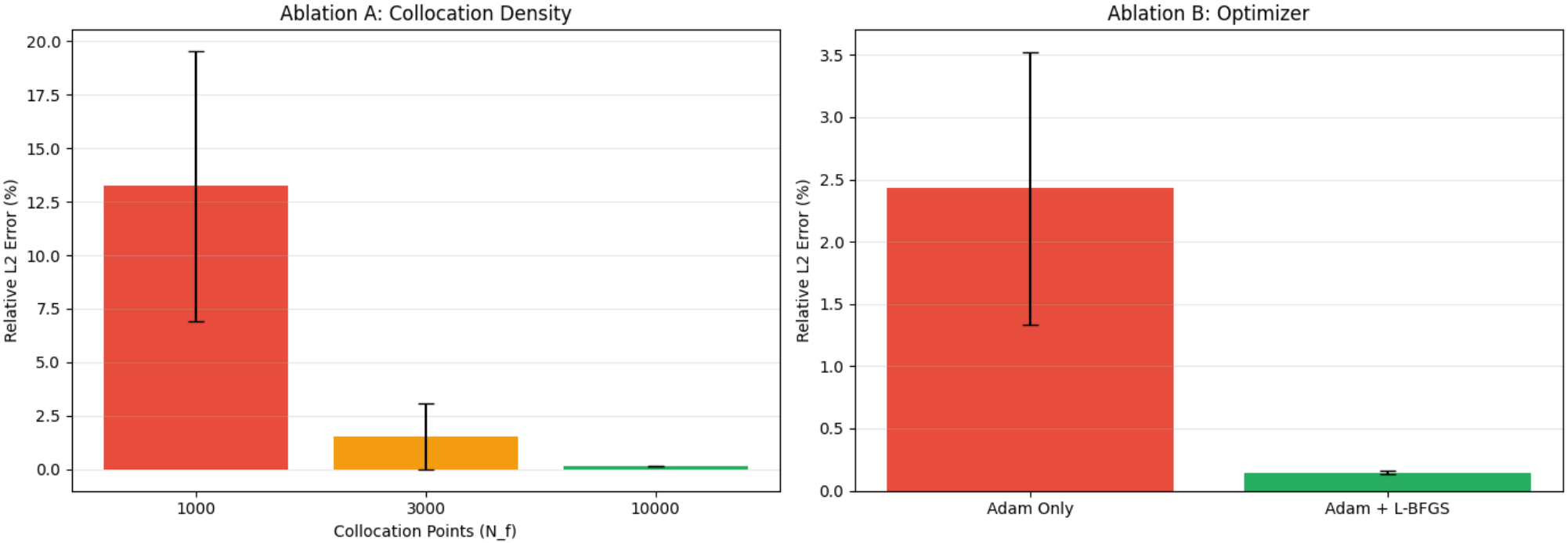
Time slices at t = 0, 0.2, 0.4, 0.6, 0.8, 1.0, showing the shock forming near x = 0.

Figure 4. Inverse problem training



Left: inverse loss (total, data, PDE). Right: viscosity estimate converging toward the true value 0.003183 (final 0.003256, 2.30% error).

Figure 5. Ablation studies



A: collocation density (N_f = 1,000 / 3,000 / 10,000). B: Adam only vs Adam + L-BFGS. Mean ± std over seeds 42, 123, 2026.